

Rahul Bainsla

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PROFESSIONAL SUMMARY

CS student (May 2027) building production systems in distributed systems, developer tooling, application security, and applied AI. Experienced with Python, TypeScript, Swift; strengths in API design, data pipelines, automated testing, and secure multi-tenant architecture.

EDUCATION

Virginia Tech

Bachelor of Science in Computer Science

Blacksburg, VA

Aug. 2023 – May 2027

EXPERIENCE

AI Intern

June 2026 – Aug. 2026

Virginia Information Technologies Agency (VITA)

Richmond, VA

- Evaluated Small Language Model integration options for public-sector workflows and authored knowledge-base documentation that accelerated responsible AI adoption and governance reviews across agency teams.
- Designed and executed end-to-end testing for the Ask Codey chatbot, identifying and reporting response-quality and reliability defects that were resolved prior to production launch.

Software Engineer Intern

Jan. 2026 – Feb. 2026

Ithena

Glen Allen, VA

- Architected a real-time Kafka telemetry pipeline integrating Isolation Forest and LSTM models, achieving a **95%** anomaly-detection rate with a **sub-5%** false-positive rate.
- Enforced exactly-once delivery, idempotent writes, and manual offset management to guarantee audit-grade data integrity across the distributed event pipeline.
- Authored alert-logic and sensor-mapping runbooks that reduced incident diagnosis time and improved cross-team knowledge transfer during production events.

Co-Founder & CTO

Oct. 2025 – Present

MedRa Robotics

San Francisco, CA

- Co-founded MedRa Robotics to develop a medical robotics platform focused on reducing clinicians' occupational radiation exposure during hospital procedures.
- Led software engineering across robotics control, computer vision, and hardware-software integration, translating clinical requirements into a functional system architecture.
- Secured a hospital pilot program and advanced the product toward manufacturing scale-up by coordinating technical development, clinical integration, and safety requirements.

PROJECTS

RepoWhisper | *SwiftUI, FastAPI, PyTorch, Faster-Whisper, LanceDB*

2026

- Designed a private macOS developer tool to solve slow code discovery across large local repositories through voice-controlled semantic search.
- Built a SwiftUI menu-bar client and FastAPI service that transcribe **16-kHz** audio, generate **384-dimensional** MiniLM embeddings, and retrieve repository-filtered code chunks from LanceDB.
- Delivered a local-first architecture with automatic backend lifecycle management, Unix domain sockets, per-install authentication tokens, and fail-closed repository allowlists.

VibeGuard | *TypeScript, Node.js, Vitest, SARIF, GitHub Actions*

2026

- Addressed security risks in AI-generated code by building a zero-config static-analysis CLI for secrets, injection, cryptography, web, cloud, and infrastructure vulnerabilities.
- Implemented asynchronous file discovery, language-aware matching, SARIF/JSON reporting, **14** automated fix strategies, and optional OpenAI, Anthropic, or Groq verification.
- Shipped **64** detection rules with GitHub Actions integration and **73** test cases covering scanning, reporting, remediation, and AI-assisted workflows.

TECHNICAL SKILLS

Languages: Python, TypeScript, Swift, SQL, JavaScript, Bash

Frameworks: FastAPI, Next.js, React, Node.js, SwiftUI, LangGraph, Vitest

Data & Infrastructure: Kafka, PostgreSQL, Supabase, LanceDB, Redis, Vercel, GitHub Actions

AI/ML & Security: PyTorch, Sentence Transformers, Faster-Whisper, Isolation Forest, LSTM, **AES-256-GCM**, Row Level Security